Questions

1. At what angle must the two forces $(x + y)$ and $(x - y)$ act so that the resultant may be $\sqrt{(x^2 + y^2)}$

(a) $\cos^{-1}\left(-\dfrac{x^2 + y^2}{2(x^2 - y^2)}\right)$ (b) $\cos^{-1}\left(-\dfrac{2(x^2 - y^2)}{x^2 + y^2}\right)$

(c) $\cos^{-1}\left(-\dfrac{x^2 + y^2}{x^2 - y^2}\right)$ (d) $\cos^{-1}\left(-\dfrac{x^2 - y^2}{x^2 + y^2}\right)$

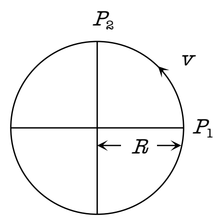
2. Figure below shows a body of mass M moving with the uniform speed on a circular path of radius, R. What is the change in acceleration in going from P_1 to P_2

(a) Zero

(b) $v^2/2R$

(c) $2v^2/R$

(d) $\dfrac{v^2}{R} \times \sqrt{2}$



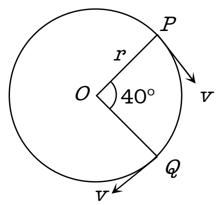
3. A particle is moving on a circular path of radius r with uniform velocity v. The change in velocity when the particle moves from P to Q is $(\angle POQ = 40°)$

(a) $2v \cos 40°$

(b) $2v \sin 40°$

(c) $2v \sin 20°$

(d) $2v \cos 20°$



4. In above example a unit vector perpendicular to both $\vec{A}$ and $\vec{B}$ will be

(a) $+\dfrac{1}{\sqrt{3}}(\hat{i} - \hat{j} - \hat{k})$ (b) $-\dfrac{1}{\sqrt{3}}(\hat{i} - \hat{j} - \hat{k})$

(c) Both (a) and (b) (d) None of these

5. For any two vectors $\vec{A}$ and $\vec{B}$, if $\vec{A}.\vec{B} = |\vec{A} \times \vec{B}|$, the magnitude of $\vec{C} = \vec{A} + \vec{B}$ is equal to

(a) $\sqrt{A^2 + B^2}$ (b) $A + B$

(c) $\sqrt{A^2 + B^2 + \dfrac{AB}{\sqrt{2}}}$ (d) $\sqrt{A^2 + B^2 + \sqrt{2} \times AB}$

6. Which of the following is the unit vector perpendicular to $\vec{A}$ and $\vec{B}$

(a) $\dfrac{\hat{A} \times \hat{B}}{AB \sin\theta}$ (b) $\dfrac{\hat{A} \times \hat{B}}{AB \cos\theta}$

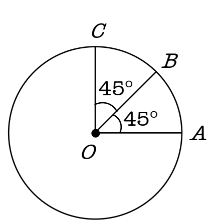
(c) $\dfrac{\vec{A} \times \vec{B}}{AB \sin\theta}$ (d) $\dfrac{\vec{A} \times \vec{B}}{AB \cos\theta}$

7. Find the resultant of three vectors $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$ shown in the following figure. Radius of the circle is R.

(a) $2R$

(b) $R(1 + \sqrt{2})$

(c) $R\sqrt{2}$



(d) $R(\sqrt{2}-1)$

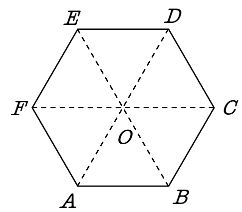
8. Figure shows $ABCDEF$ as a regular hexagon. What is the value of $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF}$

(a) $\overrightarrow{AO}$

(b) $2\overrightarrow{AO}$

(c) $4\overrightarrow{AO}$

(d) $6\overrightarrow{AO}$



9. The length of second's hand in watch is 1 *cm.* The change in velocity of its tip in 15 seconds is

(a) Zero

(b) $\frac{\pi}{30\sqrt{2}}\ cm/\sec$

(c) $\frac{\pi}{30}\ cm/\sec$

(d) $\frac{\pi\sqrt{2}}{30}\ cm/\sec$

10. A metal sphere is hung by a string fixed to a wall. The sphere is pushed away from the wall by a stick. The forces acting on the sphere are shown in the second diagram. Which of the following statements is wrong

(a) $P = W\tan\theta$

(b) $\vec{T}+\vec{P}+\vec{W}=0$

(c) $T^2 = P^2 + W^2$

(d) $T = P + W$